A Systems Engineering Based Model Selection Approach for Complex, Multi-Disciplinary Physics Problems

Efe Y. Yarbasi*, Burak Bagdatli† and Dimitri Mavris‡
Georgia Institute of Technology, Atlanta, GA, 30332, United States

Design of a complex product such as an aircraft is a long, costly, inter-disciplinary process that often take many years. Thanks to the increase in computational capabilities, engineers rely on computer models to make predictions about the expected performance of a designed product. However, the degree to which computational tools abstract the actual problem and its implications are commonly overlooked. In the absence of prior experience or validation data, it is very difficult to estimate the gap between the tool predictions and actual performance. In this paper, a systematic framework based on established Systems Engineering methods is proposed to ensure the fitness-for-purpose of possible models, keep a track of simplifying assumptions and requirements, and make a selection. The framework includes the description of the problem ontology, and how it relates to requirements, functional and physical decompositions. Furthermore, a scenario where more than one modeling option may be appropriate is used to illustrate the implications of propagated uncertainty on mission-level metrics obtained from using two different workflows for the same problem.

I. Introduction

A complex product such as an aircraft needs the pass many toll gates throughout its life-cycle, from concept initiation, to operations. The success of this product at a given point in the life-cycle is determined by how well it meets the requirements previously set by various stakeholders. A prominent example of this process is the Aircraft Structural Integrity Program (ASIP) adopted by the United States Air Force (USAF) and the Department of Defense (DoD) [1]. ASIP defines the requirements necessary to achieve structural integrity in USAF aircraft through a series of well-defined, consecutive tasks. These tasks include the requirements for any type of aircraft, for the entire product life cycle—from inception to operations, and for the entire structure. The five primary tasks of ASIP and some of the crucial sub-tasks are illustrated in Figure 1.

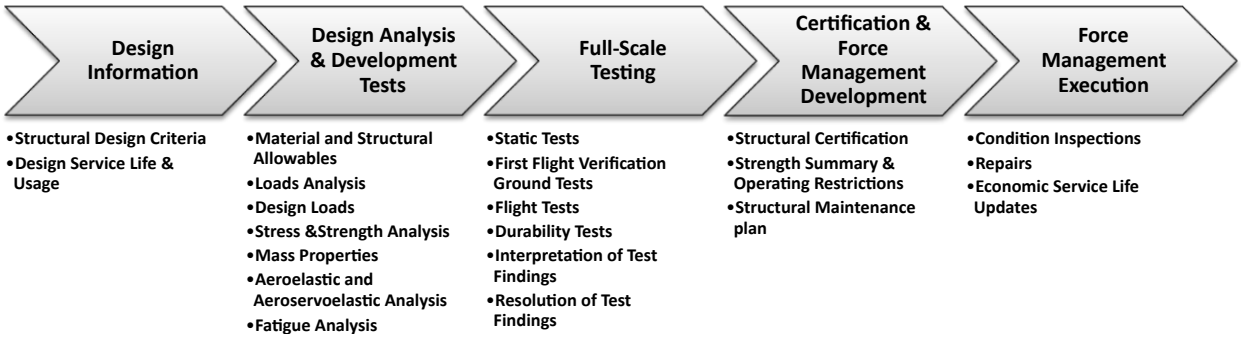


Fig. 1 ASIP Primary Tasks. Adapted from [1]

All tasks given in Figure 1 are supported by computational tools to some degree. Engineers utilize computational tools to calculate the responses of the systems they are working with. Thanks to the increase in computational capabilities in recent decades, engineers do not have to rely solely on physical experiments to gain information about systems

*Graduate Research Assistant, Aerospace Systems Design Laboratory, Daniel Guggenheim School of Aerospace Engineering, AIAA Student Member

†Research Engineer II, Aerospace Systems Design Laboratory, Daniel Guggenheim School of Aerospace Engineering, and AIAA Member

‡S.P. Langley Distinguished Regents Professor, Aerospace Systems Design Laboratory, Daniel Guggenheim School of Aerospace Engineering, AIAA Fellow

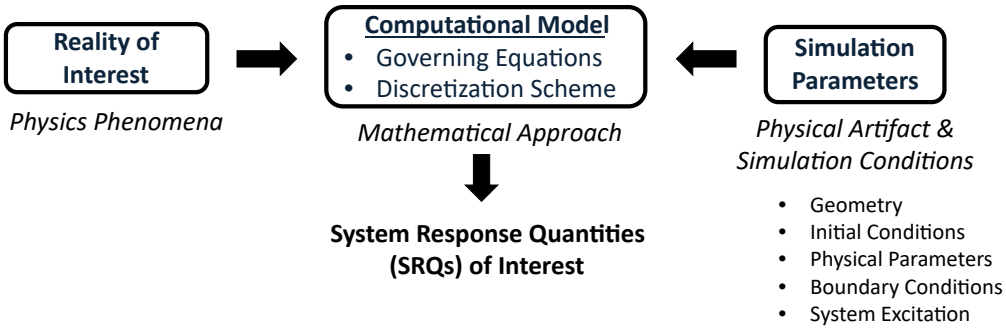


Fig. 2 Illustration of how a generic computer simulation works. Adapted from [9]

involving complex physics. Although very detailed simulations can be prohibitively expensive in terms of work-hours and computation time, the use of computer simulations is still cheaper and has a quicker turnaround time compared to following a build-and-break approach; especially for large project.

In the earlier design stages of an aircraft, where the design is not fixed yet; computational models immensely help in making risk-informed decisions. As the predictions obtained from simulations played a larger part in the design, concepts such as "*simulation driven design*" has been more prominent in many disciplines. Although this concept is perceived differently by subject matter experts in various domains, a good overview of the state-of-the-art can be found in Reference [2]. Moreover, the use of computational models are not limited to design only. They can still be used for support activities well after the design as given in Figure 1; such as certification, maintenance and inspections . Having highly accurate computational tools has the potential of bypassing some or all of the physical tests necessary in order to certify the aircraft. Because physical tests are very expensive and long campaigns [3], replacing them with virtual counterparts can potentially save vast amounts of time and money [4, 5]. Certifying* aircraft parts by using computational tools only is called *Certification by Analysis* (CbA), and is a long-term goal of stakeholders such as the FAA (Federal Aviation Administration) [3]. Furthermore, incorporation of information obtained from sensory networks enable creation of *digital twins* of an aircraft, which can be used for tasks during operational life such as predictive maintenance [6, 7].

A. Modeling Physical Phenomena

Computational tools are almost exclusively used to make predictions about the response of a system under a set of inputs, boundary conditions, and external excitations. At the core of these tools lies a *model*, which represents the reality of interest, commonly in the form of mathematical equations (e.g., a set of integro-differential equations) derived from previously observed phenomena and measured data [8]. If an explicit analytical solution is infeasible under computational capabilities at hand, or prone to divergence, the solution domain can be discretized in the form of *grids* so that the analytical solution can be approximated. The outputs of a tool are frequently called *System Response Quantities* (SRQ) or *Quantities of Interest* (QoI). How a generic computer simulation works is illustrated in a general sense in Figure 2.

When it comes the modeling physical processes, almost all models are abstractions of the reality of interest as they do not consider the problem at hand in its entirety, but only general aspects of it that yield useful conclusions without spending time on details that do not significantly impact the information output. This is very much in line with the famous aphorism: "*all models are wrong, but some are useful*" [10].

Generally speaking, models that abstract fewer details of the problem are able to provide more detailed information with a better accuracy, but they also require more detailed information (i.e., high number of model parameters) about the system of interest and external conditions to work with. Such models are called high-fidelity models. Conversely, low-fidelity models abstract larger chunks of the problem and have a quick turnaround time in terms of model building and calculation. They may be able to provide valuable information without so much effort going into setting up the model; unlike high-fidelity models, they can do it with very low computational cost. Especially in time-constricted endeavors, the selection of the correct model that has the correct capabilities is a task that needs to be tackled meticulously.

Consider an array of models that attempt to represent the same problem in different levels of detail. When these

**Qualification* for military applications

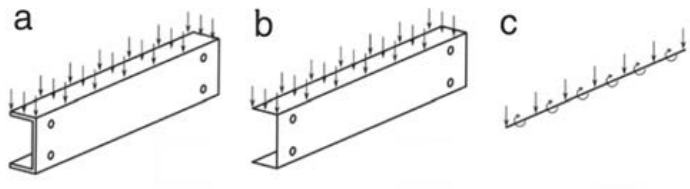


Fig. 3 Abstraction examples of forces acting on a beam. a. Three-dimensional solid body representation b. Mid-surface representation c. Distributed edge load and moment on 1-D center line representation. Taken from [12]

models are sorted according to their fidelity levels, there will be an emergent physics phenomenon to be captured going from one level to another. An example abstraction is given in Figure 3 in the form of how the representation of the load distribution changes as the geometry is simplified. Such differences in fidelity will impact how phenomena such as bending and shear effects are modeled in different levels. The abstraction in geometry and physics will have an impact on the desired response as well as the the decision of whether the model is adequate for its intended use. When making abstractions, one needs to be careful as the analysis capability can mask the real behavior [11].

There might be important differences in how the problem is represented, especially the media of interest and boundary conditions. Consider modeling of a gas moving over a surface, the medium can be modeled as a bulk (continuum) or at a molecular level. Corresponding assumptions regarding the medium (e.g., ideal gas, real gas effects for air; material homogeneity for solid modeling) play an important role in deciding how much detail is required to be captured for an intended purpose. Furthermore, the behavior of the medium (e.g., elasticity, compressibility) may have a great impact on the outputs of a computational tool. This is predominantly governed by how the governing equations (laws of physics) are captured in the models in the form of mathematical equations. The number of terms and coefficients in these equations, as well as the physical and geometric properties of the domain over which the equations are solved, impact the number of options one has to represent the problem.

Another important aspect of physics-based models is the boundary conditions. This defines the *external* boundaries of the problem and how the modeled artifact interacts with its surroundings. This can typically be an outer mold line (OML) for CFD applications, or surfaces/points of contact for computational structures modeling (CSM). Similar to the medium, the level of detail in modeling the boundary conditions is also crucial. For example, sophisticated internal modeling may be necessary for modeling of composites, additive manufacturing, whereas simpler 1-D, steady boundary conditions might suffice for more trivial beam models. In engineering applications, boundary conditions are frequently dictated by functional or performance requirements (or requirements derived from these), such as altitude (density) or Mach number (compressibility, governs shocks), or they may directly depend on external models or industry standards, such as standard atmosphere or material properties.

B. Sources of Uncertainty

There are many sources of uncertainty involved in the process of designing a complex, multi-disciplinary product, such as an aircraft. In such long programs, it is of utmost importance to identify these sources, classify them accordingly and follow a consistent terminology throughout the program between all involved parties, so that they can be properly addressed. Researchers have come up with several taxonomies [13–15], depending on their area of application and their preferences. There cannot be a single correct uncertainty taxonomy that would work the best for all disciplines. However, a taxonomy that is consistent with its intended use will eliminate possible ambiguity and enable correct characterization of various uncertainty sources.

From a modeling perspective, uncertainties are the reason of the discrepancy between simulated SRQs and their real-world counterparts. Two types of uncertainty that one can commonly found in literature are: *epistemic* and *aleatory* uncertainty. The word "epistemic" is derived from the Greek word for "knowledge". This type of uncertainty is due to a lack of knowledge [8, 16]. In theory, it is reducible by obtaining more information about the system in question. Therefore, it is also referred to as *reducible uncertainty*, *de dicto*, and *Type B* [8, 17–19]. The word "aleatory" is derived from the Latin word *alea*[†] (eng. *a die*). Aleatory uncertainty is attributed to the inherent randomness in the problem [8].

[†] *"Alea iacta est!"* (eng. *"The die is cast!"*) is a famous phrase attributed to Julius Caesar. It is believed that he uttered these words as he led his armies across the Rubicon. Ironically, it is more probable that he gave life to this phrase in Greek, not in Latin [20].

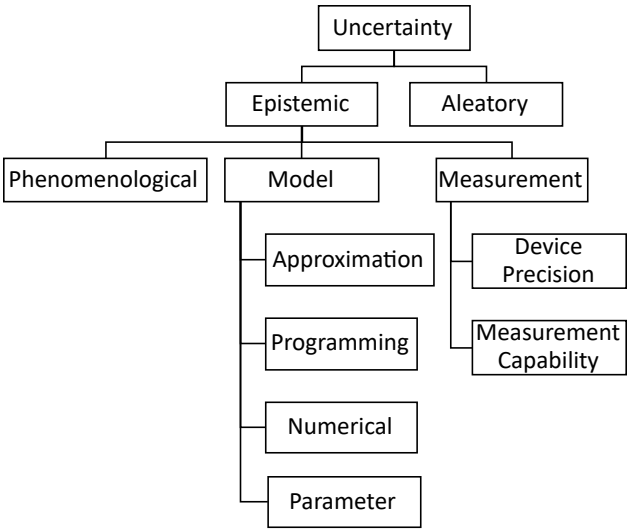


Fig. 4 Uncertainty taxonomy used modeling and simulation purposes

Therefore, it is also referred to as *stochastic*, *random*, *de re*, or *Type A* [8, 16, 19, 21]. Oberkampf et al. [15] proposed an uncertainty taxonomy created for modeling and simulation purposes that is consistent with aforementioned uncertainty types. In this taxonomy, errors are distinguished from the uncertainties. Furthermore, errors are divided into two categories: *acknowledged* errors such as modeling approximations, and *unacknowledged* errors, such as computer programming mistakes.

Reviewing the uncertainty taxonomies in literature, the following taxonomy given in Figure 4 is proposed for modeling and simulation purposes. It contains the minimum set of uncertainty types pertaining to modeling and simulation activities. This taxonomy follows the classic divide of epistemic and aleatory uncertainties. Epistemic uncertainty is further divided into two parts. Phenomenological uncertainty is due to a lack of knowledge in the physical phenomenon that is in question. If the phenomenon is not fully understood, or some information about it is unavailable, it will introduce uncertainty in the decision-making process. This is also sometimes referred to as "unknown unknowns" [22]. Only subjective approaches can be used in characterizing this type of uncertainty as there is no a priori information available. Model uncertainty is simply related to the construction and execution of a model in question. Programming and numerical errors refer to the mistakes in coding and solution algorithm, or parts of the solution, respectively. *Approximation errors* refer to the approximation of the problem with respect to the reality that the model intends to capture. It has nothing to do with a mathematical nature. It is actually an *acknowledged* error and defined also as *model-form* error [8]. Parameter uncertainty is present due to the fact that the exact value of parameters in simulations such as coefficients, inputs, boundary conditions that would perfectly reflect the reality are almost always unknown.

In almost all engineering applications, engineers employ some kind of modeling and simulation environment to justify their decision making. Therefore, it is important to understand the sources of uncertainty in the task of model selection and evaluation, and what the potential implications are. Changing the level of fidelity of the model will have an impact on most of the listed uncertainty types. Therefore, one needs to make a careful assessment between uncertainties involved in different types of models. In this paper, the main goal is to address how model related uncertainties (primarily approximation and parameter) impact the model selection process and to investigate the outcomes.

C. Multi-Disciplinary Analyses

Most problems in the domain of aerospace engineering are very complex and often multi-disciplinary. Therefore, holistic approaches are often infeasible and the use of highly accurate methods is limited to a single or few design points [23]. Therefore, connecting different disciplinary tools to model a problem is a common practice. Because of its importance in aircraft design and operations, aero-structures analyses of an aircraft wing will be used to demonstrate the process throughout this paper.

One of the most important physics phenomena in the case of flight vehicles is the interaction of aerodynamic loads

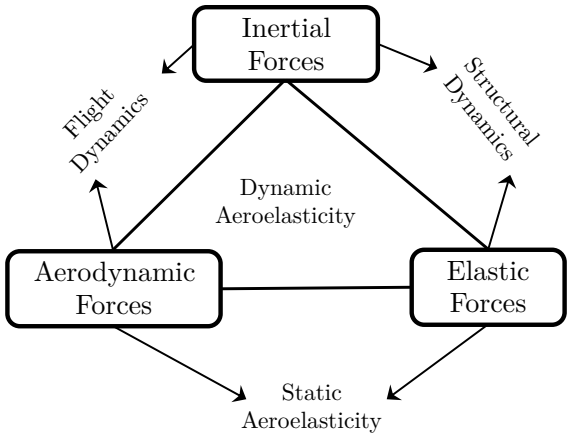


Fig. 5 Forces that make up an aeroelastic phenomenon, Collar's Triangle. Adapted from [24]

with structural displacements, i.e., aeroelastic phenomena. The impact of this inter-dependency on performance and reliability have been investigated since the early days of aeronautics. From a simplistic standpoint, the differential equation that represents the motion of an elastic body under aerodynamic loads can be written as:

$$M\ddot{x} + Kx = P(x,t) \tag{1}$$

where x is the position vector, M is the mass matrix, K is the stiffness matrix, and $P(x,t)$ is the aerodynamic loads as a function of position and time. The first term in the equation represents the inertial forces arising due to the acceleration of the mass structure, whereas the second term represents the elastic forces due to bending and compression of the material. Figure 5 shows the different classifications of disciplines as the interaction of these forces, as identified by Collar [24]. A phenomenon that can be classified as a *dynamic aeroelastic phenomenon* occurs due to a significant interaction of all three of these forces. Some of the most prominent aeroelastic phenomena are because of the wings deforming under aerodynamic loads. Certain conditions may trigger various failure modes such as divergence and flutter. Depending on the physical artifact and flight conditions, more phenomena can be listed.

D. Multi-Fidelity Analyses

As discussed before, the advancements in computation enabled the analysts to develop high-fidelity codes that are capable of making accurate and detailed predictions regarding a complicated physics phenomenon. However, this is a blessing and a curse. Such tools, such as CFD, require very detailed information regarding the artifact to be modeled, and even then, they are prone to convergence issues [25]. The latter can be addressed as soon as the model is developed and the results are observed to be inconsistent, but the former must be questioned even before the modeling exercise is initiated. Especially in the design of a product such as an aircraft, such details are simply not available. Therefore, attempting to develop a generalized high-fidelity model without the required information is not possible. Even if was, it would still be subject to convergence issues and prohibitively expensive computational burden for exploratory, parametric analyses required for evaluation of alternatives.

Multi-fidelity analyses aim to leverage the low cost computation of low-fidelity tools and detailed predictions of higher-fidelity tools. Although the use of multi-fidelity models is a broad area where the execution philosophy, number of tools in an ensemble, sampling methods, the set of assumptions and the computation methodology can vary greatly between different problems, depending on the specific constraints of the problem and the preferences of the analysts, they constitute the perfect test bed in order to demonstrate the proposed framework in this paper. While this means the comparison of such tools will enable selection of a tool that is *fit-for-purpose*, it also means the best tool for the job may be some kind of a coupling of different analyses.

One of the most important questions in multi-fidelity methods is how high-fidelity results and run time can be balanced with the practicality of the low-fidelity analyses. One such example is Xie et al. [26] using a multi-fidelity framework to estimate the aerodynamic characteristics of a Hybrid-Wing-Body (HWB) aircraft and compare the obtained results with the available data for the ONERA M6 transonic wing. Their analysis incorporates a 3-D Vortex Lattice Method (VLM) as the core tool for drag calculation, but it is calibrated with drag coefficients obtained from a surrogate

model created from 2-D CFD airfoil results. In this work, the higher-fidelity tool is seen as a one-time investment and need not be online in calculation. If they can be trusted and certain validation data exists, they can be used to calibrate the results obtained from lower-fidelity tools. However, it can also be an online calculation where the high-fidelity tool is queried whenever necessary. A prominent example is Reference 27, where three different levels of fidelity (ASTROS, ZEUS, and inviscid FUN3D) are used for aerodynamic calculations, whereas structural calculations are handled by a single solver. They calculate the critical flutter dynamic pressures for the AGARD 445.6 aeroelastic test case as a function of angle of attack, Mach number and thickness-to-chord ratio, as some validation data obtained from physical measurements exist for this wing. They note, using three different fidelity models create more accurate results compared to the high-fidelity model alone, as well as savings in computational time. Obviously, creating multiple models and ensuring that the results are transferable and the framework works as intended will take more time and effort than using any one model, but the convergence, uncertainty and code-to-code validation benefits may worth the effort; especially for complex simulations that take a long time to run.

Although the use of multi-fidelity analyses are more common for the calculation of aerodynamic properties, structural calculations can be carried out with with the same mindset. Solution methodologies, as well as geometric abstractions can be treated as different fidelity levels. Yong et al. [28] present the comparisons of solid mesh studies versus medial mesh studies in the context of structural optimization of an engine. They also note the effectiveness of using multi-fidelity simulations in terms of accuracy at a relatively low additional effort. Diazdelao [29] shows that multi-fidelity methods (a Bayesian method based on Gaussian Process emulators in this case) work on non-convex geometries to approximate Finite-Element methods at a lower computational cost.

E. Systems Engineering Perspective

Because the problems in aerospace engineering are often very complex, practitioners need to break down the problem into smaller bits, carry their analyses at a lower level and integrate later. Therefore, Systems Engineering principles are commonly adopted in the aerospace industry. The end-goal of reduced reliance on physical testing during development campaigns increased the reliance on computational models and incorporation of Systems Engineering practices. This paradigm shift caused established engineering processes to be re-considered. As an answer, concepts such as Model-Based Engineering (MBE) and Model-Based Systems Engineering (MBSE) have emerged in the 70s have been recently gaining popularity [30]. MBE and MBSE use systems Verification and Validation (V&V) techniques to support model development process, informed by the system requirements for system tests and evaluation. In the US, the DoD is spearheading change towards wider adoption of MBE and MBSE throughout the industry [31]. Corporate and research strategies are adopting accordingly in anticipation of a future in which: "*Over the product lifecycle, increasing detail and fidelity will be added to virtual products, production systems, and product support ecosystems to enhance the quality, cost, performance and safety of their physical counterparts. But more important, the data will be connected through models and simulations to provide a single source of information that is authoritative, live, and consumable*" [32].

Recent increases in computational capabilities and consequently the role of computers in the design of complex systems, has paved the way for established methods to adapt. A prominent example is the Boeing's Model-Based Engineering Diamond. [‡] In this approach, the bottom part is the classical system V [33], whereas at the top, there is another part that mirrors the "physical V" in a virtual domain. The two parts of the diamond are connected to each other by a *digital thread* which links models and simulations to the product design. Virtual and physical parts inform each other throughout the life cycle of the product. Although, approaches such as the Boeing Diamond do carefully tie modeling and simulation tools to system design together, it does not have an emphasis on what attributes a modeling and simulation environment *should have*. Fitness-for-purpose of the modeling environments is a missing link in activities throughout the product life-cycle. Similar to V&V activities performed on the product, established Systems Engineering principles can be invoked for the process of selecting or developing a computations tool that models an engineering problem. Balci [34] defines that the requirements imposed on a modeling and simulation environment need to following quality indicators: *accuracy, verity, validity, clarity, completeness, consistency, feasibility, modifiability, stability, testability* and *traceability*.

A modeling and simulation environment can be thought of as a product, as it is developed, checked for requirement compliance, and used for a desired purpose. It can be an overall M&S application or a work product during its development [35]. Similar to a product, the interdependencies between its components, the integration scheme should be clearly expressed. In multi-disciplinary analysis and optimization (MDAO) applications, a Design Structure Matrix

[‡] frequently referred to as "Boeing Diamond"

(DSM) is traditionally used to convey the information regarding how the workflow is set up. However, they typically do not depict crucial information, such as what variables are passed between analyses, whether there is a loop or not, and the details of coupling between analyses. These issues have been addressed by developing an *Extended* Design Structure Matrix (xDSM). xDSM visualizations are designed to make the interfaces clear between the MDA components and specify the algorithm by which the MDA problems are solved [36].

II. Proposed Approach

A. Systems Engineering in Uncertainty Identification

The framework described in this section aims to develop a Systems-Engineering guided model development/selection approach that will enable identifying of important sources of uncertainty. In this section, the elements of the framework will be described and the steps will be followed on the example of the design of a military mobility transport aircraft.

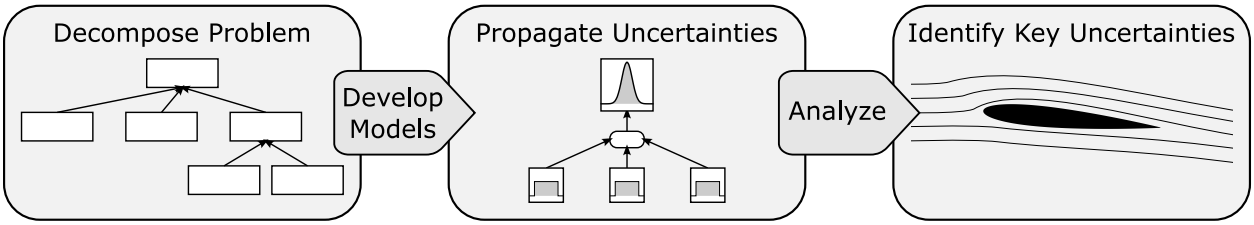


Fig. 6 Systems engineering process for identifying critical uncertainties in design [37].

At the first stages of the design of an aircraft, there is great freedom on what specifications that the final product may possess. As more decisions are made, the design is locked in. In other words, the knowledge about the design increases as the design freedom decreases [38].

B. Formulating the Problem Ontology

The proposed approach requires carefully describing the problem ontology[§] so that a modeling and simulation environment (MSE) that best represents the problem at hand can be created. The premise here is that a modeling and simulation environment built by following requirements, functional, physical decompositions and their mapping onto a modeling architecture will enable tracking of model borne uncertainties. This step corresponds to the problem decomposition task illustrated in Figure 6 and an overview is given in Figure 7. Each decomposition should enable keeping a list of various types of abstractions and sets of assumptions in their own respective domains. The union of processes within the a dashed line is called the *problem ontology*.

This approach will include mapping of requirements of different levels to be mapped onto different decompositions and also the modeling architecture so that the modeling and simulation environment can be created or selected according to higher-level requirements. The model inputs, boundary and external conditions will be provided from functional and physical decompositions. Depending on the artifact, details about the modeled geometry and related physical abstractions will be listed and kept track of throughout the process. Modeling architecture includes the domain-specific physical phenomena, list of modeling assumptions, and also cross-domain properties.

All aircraft are designed to answer a specific set of requirements that are borne out of needs of the market or stakeholders. Definition of the *concept of operations* outline the broad range of missions and operations that this aircraft will be used for. After the overall purpose of the aircraft is determined, the designers can decide on the required capabilities and the metrics that these capabilities are going to be measured with can be derived. Considering these information, a concept of the aircraft and the rough size of the aircraft can be determined. This process can be completed by *decomposing the requirements* and tracking their impact on capabilities and metrics. Some aircraft may have a multi-role nature in their design. In those cases, additional capabilities will emerge and parallel decompositions may need to be used. A multi-role aircraft will almost always have a more stringent set of requirements compared to a

[§]The definition used in this context is closer to what has been defined as a *computational ontology: "Computational ontologies are a means to formally model the structure of a system, i.e., the relevant entities and relations that emerge from its observation, and which are useful to our purposes. An example of such a system can be a company with all its employees and their interrelationships." [39]*

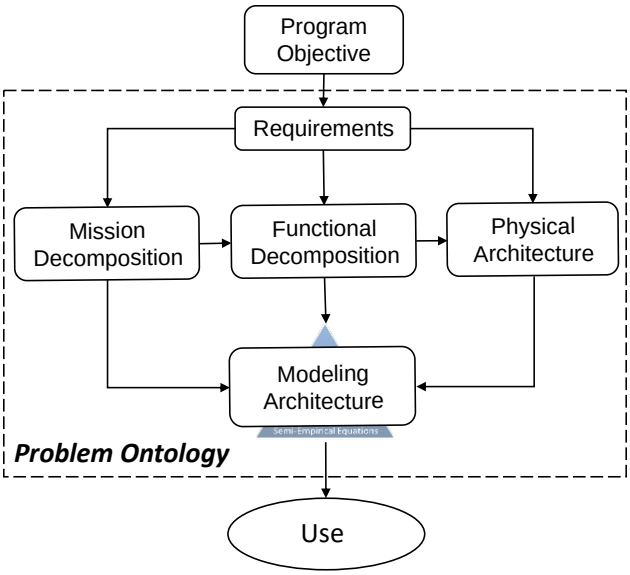


Fig. 7 Problem Ontology

comparable single-objective aircraft. In order to demonstrate the process, the development of aero-structures analysis capabilities in the early stages of the design of a military mobility transport aircraft has been selected.

1. Requirements Decomposition

An example *requirements decomposition* is shown in Figure 8 for a military mobility mission. For this operation, important capabilities may be listed as speed, range, payload, access and reliability etc. Metrics, such as the ones given in Figure 8 determine the concept that will have to satisfy the requirements. For example, a military transport aircraft that will have easy access for loading and unloading of large cargo with a 5500 nmi range will have to be a high-wing configuration large aspect ratio wing planform. From a conceptual design perspective, there might be multiple concepts that would meet aforementioned requirements. For the given example, a traditional configuration such as a Lockheed Martin C-5 Galaxy or novel configurations such as a Hybrid-Wing-Body (HWB) can be suitable candidates, because they would satisfy the capability requirements. Preference of a concept over others can be left to a later stage, if a strong dominance due to an attribute or budget concerns does not exist at the current stage.

2. Mission Decomposition

After high-level system requirements are decomposed and corresponding capabilities and metrics are identified, the analysts can scrutinize the design mission. For example, a generic mission for a military mobility aircraft will include segments such as taxi, take-off, cruise, landing. At each segment, there will be another set of requirements allocated specifically to that segment. These metrics can be further linked to metrics that need to be tested in a modeling and simulation environment. These decompositions will help in mapping the analysis scenario the the simulation inputs and parameters. An example mission decomposition is illustrated in Figure 9. The task at hand is the modeling and simulation of the aircraft wing for the mission, the most relevant metric given in Figure 9 will be the maximum load factor that the structure needs to sustain during a maneuver, which is 2.25g.

3. Functional Decomposition

Functional architecture is defined as an "*arrangement of functions and their subfunctions and interfaces (internal and external) that defines the execution sequencing, conditions for control or data flow, and the performance requirements to satisfy the requirements baseline*" [40]. Included functions can be, for example, performance requirements, physical requirements, etc. Depending on the modeling task, the functional architecture should follow the life-cycle evolution of the system. For example, early on in the design process, high-level functions regarding the mission phases are covered, but later, domain-based functions, detailed interface functions, manufactured interface functions and operational

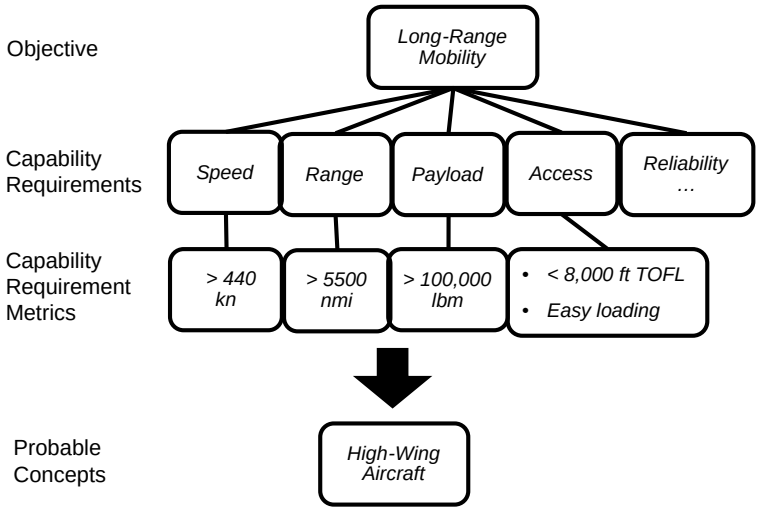


Fig. 8 Decomposition of design requirements for a mobility transport - Capability Perspective

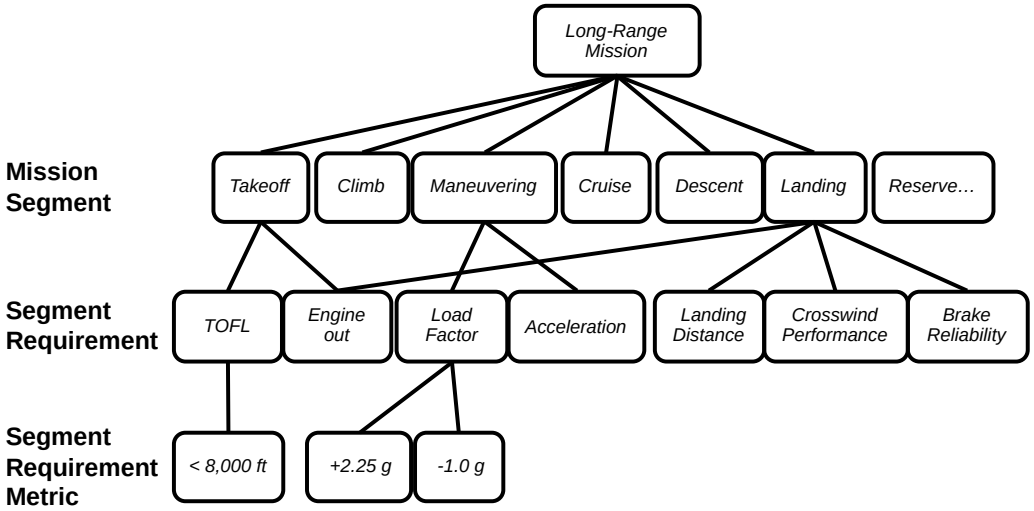


Fig. 9 Decomposition of design requirements for a mobility transport - Operations Perspective

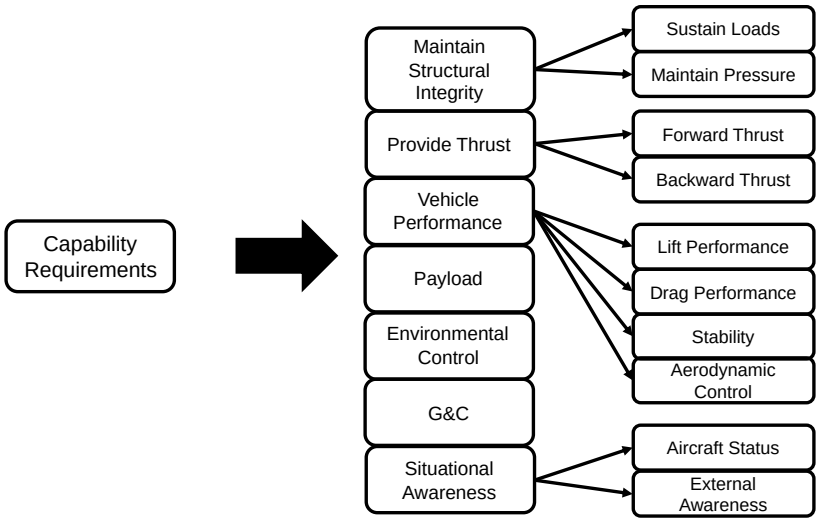


Fig. 10 Functional Decomposition. Adapted from [41]

requirements will come into play. The models used in this evolution will adapt or change entirely.

Decomposing the functional architecture into smaller functions helps to isolate the functions that are necessary to be included in the modeling and simulation environment, and map them to corresponding sub-assemblies in the physical architecture. The main functions are determined by the concept or concepts (family of architectural choices) obtained from program objectives, requirements decomposition and previously defined/derived requirements. Every modeling task involved in the product development will have different decompositions to fit their goals. Figure 10 shows a possible functional decomposition for a conceptually defined military cargo transport. The middle column represents all the various functions that the aircraft needs to be capable of, and the right column shows main functions decomposed into disciplines. These disciplinary functions will determine the simulation or experiment conditions as well as what sub-assembly that must be used in such a scenario. Depending on the application, the number of columns can change and other functions can be included.

4. Physical Decomposition

Physical architecture is defined as "*an arrangement of physical elements (system elements and physical interfaces) which provides the design solution for a product and is intended to satisfy logical architecture elements and system requirements*" [42]. Similar to functional architecture, it will also have a life-cycle evolution. At the early stages, where the details about the final product are not certain, it will only include the anatomy and layout. As time progresses, components in CAD drawings, bills of materials, parts for assembly, and even the components required for operational life, such as maintenance, can be included. The detail of physical architecture and the number of levels and branches of the decomposition will change depending on the life cycle of the product.

Physical architecture shows the decomposition of physical systems and their constituent physical parts. Constructing a physical architecture is crucial because it highlights what sub-assemblies need to be included in the corresponding modeling and simulation environment. Another necessity of having a physical architecture is that parts have different critical conditions that lead to design requirements. An accurate mapping between functions and physical components makes the job of engineers easier and more traceable. If the task is to design a new aircraft from scratch, and there exists more than one concept that answers the same set of program goals, there might be parallel physical architectures that correspond to different concepts, such as a hybrid-wing body (HWB) or a conventional configuration aircraft. An example physical architecture is given in Figure 11. Depending on the simulation purpose, modeled sub-system, and fidelity requirements; a necessary sub-assembly will be selected from the architecture. The level of abstraction (fidelity) will describe the level to which the geometry is abstracted, and therefore the sub-assembly that is included in the M&S environment. Similar to the functional architecture, the number of layers that define an architecture is completely arbitrary and depends on the problem. There will be no one correct way to group interacting physical components.

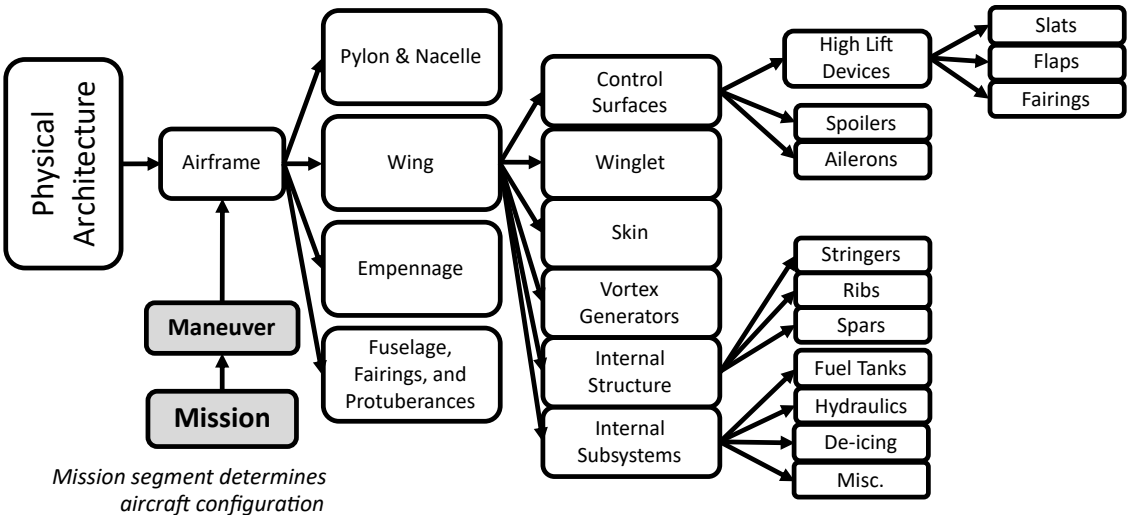


Fig. 11 Physical Decomposition

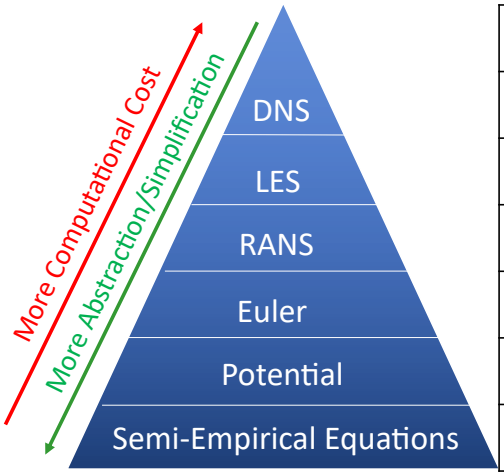
5. Modeling Decomposition

A crucial aspect that needs to be taken into consideration in the analysis of multi-disciplinary physics phenomena is the computational tools that are to be used to model them, and their place in the life cycle of the product. Using the available tools without proper verification of whether they are fit-for-purpose may result in overlooking inherent assumptions and abstractions built in the tools, or skew the results in an untraceable fashion.

In earlier design stages, there is a lack of detailed information about the design [38], which makes using higher fidelity tools may be a waste of time and effort due to the sheer number of crude assumptions, or simply the lack of detailed information pertaining to the required high number of simulation parameters. Therefore, complex problems are generally simplified at the earlier stages. Early, inherent abstractions manifest themselves as parameter or model-form uncertainties. Furthermore, a particular system needs to be modeled for the conditions that they will operate under. This necessitates a modeling and simulation environment to be *scenario-driven*. If the it needs be modeled for a multi-disciplinary conditions, a Multi-Disciplinary Analysis (MDA) environment will need to be created to obtain simulation result. In aerospace industry, this is often the case because most computer tools are in fact, disciplinary. An MDA environment involves the same system being modeled in different disciplinary tools, according to their own set of modeling assumptions and abstractions, and communicate each other's result the the other to converge on a stable result. The use case used in this work, the aero-structural analysis of an aircraft wing, is such a case. Aerodynamic forces (loads) are usually calculated by using an aerodynamics tool. This tool would require the deformed shape of the aerodynamic surface to predict the loads for that deformed geometry. Elastic and inertial forces and the deformed geometry are typically calculated using a structural analysis tool. The iteration between such disciplinary tools and the information flow is what constitutes the MDA.

In order to model a single problem, there is often more than one computational tool available. An example in the form of aerodynamic analysis tools is given in Figure 12. The triangle demonstrates a hierarchy of tools that can be modeled. The levels in this triangle, from the lowest-fidelity to the highest are: semi-empirical equations, potential flow, Euler methods, Reynolds Averaged Navier-Stokes (RANS), Large Eddy Simulation (LES), and Direct Numerical Simulation (DNS). These do not form an exhaustive list by no means. In the given order, a modeler has more capability to represent the problem by abstracting fewer aspects of the system. Although they are able to reduce the model-form uncertainty, this comes at a higher computational cost. Each fidelity level comes with their own set of assumptions and parameters. Some of the main features of each level are listed in the table given in Figure 12. In building such hierarchies, one needs to be careful as categorization is not very straightforward. In order to make a certain fidelity level to capture additional features, add-on modules can be added. Such modules enable to extend the model capabilities without drastically changing the inherent set of assumptions at that level. Furthermore, empiricism[¶] is often incorporated at all levels, but it is more prominent in the lower-levels as these models are generally constructed with a more broad set

[¶]"*the view that all concepts originate in experience, that all concepts are about or applicable to things that can be experienced, or that all rationally acceptable beliefs or propositions are justifiable or knowable only through experience*" [44]



Assumptions	Captures
Full N-S Equations	All continuum properties
Dynamic, scale-resolved	Unsteady coherent structures
Mean viscous flow	Boundary layers, some turbulence behavior
Inviscid, no heat conduction	Rotational Flow, shocks
Irrotational, isentropic	Potential Flow
General trends	Simplified behavior based on exp. data

Fig. 12 Aerodynamic modeling hierarchy. Adapted from [43]

Table 1 Structures modeling hierarchy. Adapted from [45]

			Assumptions	Captures
Structures Models	Continuous	Beam Dynamic Eqn's	• Continuous solutions • Homogeneous material	• Simple deformations and stresses
	Discrete	Beam FE	• Slender beam elements to model 3-D structures	• Deformations and stresses on typically long structres
		Shell FE	• Complex structures • No in-plane stresses • Suitable for thin structures	• Surface behaviour on complex structures
		Complex FE	• 3-D solid or mixed meshes	• All stress and strain behaviour *may* be captured

of assumptions.

Similar to aerodynamic models, modeling the structures can be performed in many different ways. If the structure can be appropriately represented as a beam, the equations of motions obtained from different beam theories (e.g., Euler-Bernoulli, Raleigh, Timoshenko) can be used as a broad abstraction. Such models employ heavily simplifying assumptions such as slender beam, no rotary inertia, no torsion or no planar warping etc. These differential equations can be further discretized to algebraic equations using techniques like modal superposition [45]. Such models heavily rely on geometric abstraction, and possess no way of modeling complex structures such as an aircraft wing-box in detail. If more geometric detail is available, Finite-Element (FE) models are commonly used. Depending on the the availability of details, rigor, and computational power; analysts can opt to utilize discretized beams, shells (2-D plates), or solid meshes to model a complex structure. A broad grouping of aforementioned structures modeling methods are illustrated in Table 1. Although this classification is not meant to be form an exhaustive list, it provides an overview on the plethora of available tools.

An example modeling decomposition is given in Figure 13. This decomposition is formed as a result of previously discussed decompositions and it has different levels by itself. How they are related were previously shown in Figure 7. The physical architecture (concept) represents the system that is to be modeled in these analyses. For every physical part that is to be simulated, a different M&S environment, therefore, another modeling architecture can be created. In order to create the model, relevant disciplines must be identified. For the sake of clarity, boxes related to aerodynamics are

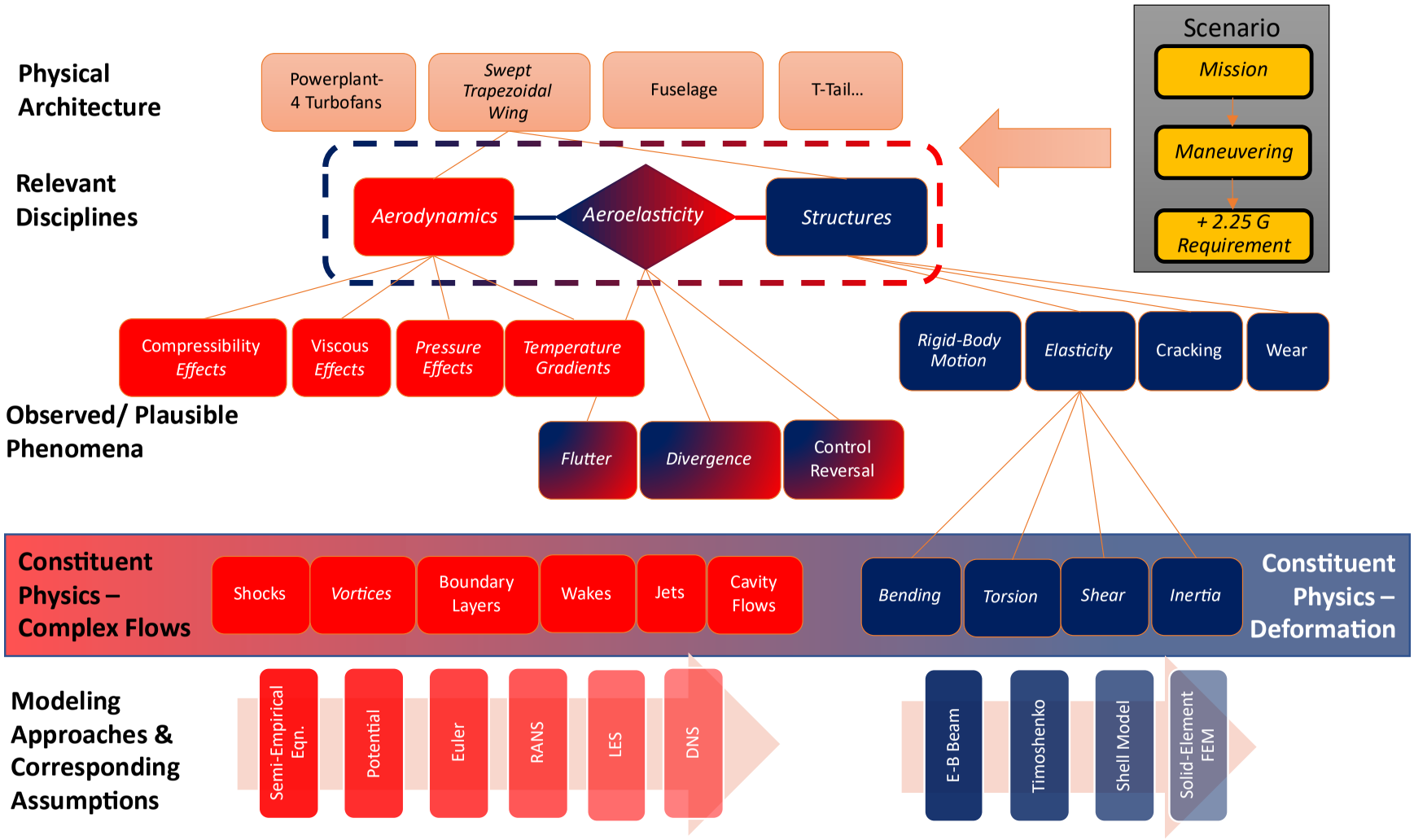


Fig. 13 An example modeling decomposition for aero-structural analyses

colored in red and structural analysis are colored in blue in Figure 13. Multidisciplinary aspects of the problem (i.e., aero-structres analyses, aeroelasticity) are illustrated with a color gradient. The phenomena that are either observed in similar problems or plausible/expected ones based on subject matter expertise are identified. These physics phenomena can be identify by employing different methods. One method that is commonly used in simulation and nuclear power communities is a phenomena identification and ranking table (PIRT) [46, 47]. It has been used for V&V and uncertainty quantification purposes for more than two decades. It enables the engineers to prioritize the some of the physics phenomena for their impact on the overall simulation and V&V activities are carried on further. After the phenomena to be investigated are identified, the constituent physics that make up these phenomena are listed and ranked. Such methods assume are based on sufficient knowledge about the physics and are may not be suitable for phenomena exploration purposes. It is crucial to be careful of where these physics phenomena occur on the body and how they are to be modeled by a computational tool. As a reminder, the tools are given in the bottom row of Figure 13 in the order of increasing fidelity, similar to the aerodynamic modeling options given in Figure 12, and the structures options given in Table 1.

C. The Decision Making Process

Noting all the discussion above, the selection of the best tool for the job at hand will require following a systematic procedure that includes establishing the value, defining the alternatives in each tool and then, making a selection. This whole process can be treated as a product design process. The following steps will be based on Integrated Product Process Design (IPPD) [48]:

1. Establish the need:

A task of modeling and simulation begins with the need. In the case of a large corporation, there might even be already available tools that can accomplish the task, this whole process can be bypassed. The analysts must do their due diligence to understand what they are going to do with the tool and what they are going to try to accomplish. At this step, the top-level requirements on the M&S environment need to be set.

2. Define the problem:

The most critical problem for selecting or developing a physics-based modeling and simulation environment is about understanding what the capabilities and limitations of a certain M&S environment will be, as it will be very hard to change later on in the product life cycle. This step will involve reviewing the literature, exploring previous attempts at solving the same or a similar problem, understanding the physics in detail. If the physics is too complicated or new (out-of-reach for current computational capabilities), then this process must be learning-based. In other words, any new knowledge obtained must be incorporated in state-of-the-art modeling and simulation tools so that phenomenological uncertainty can be minimized.

If the physics are fairly well understood, the necessary level of detail and required modeling capabilities can be substantiated by using methods such as PIRT or validation hierarchies [49]. These structures will enable the analysts to understand the fidelity of standalone physics (or an interaction of them) that the computational tool need to handle. If the problem is multi-disciplinary, then these steps need to be repeated for each discipline.

The decompositions that were discussed in Section II.B and illustrated in Figure 7 will be immensely beneficial at this step. Following a systems engineering based methodology, engineers can track the requirements and how they impact the problem at hand at different levels during this process. The decompositions will enable mapping the product and/or simulation requirements onto the modeling requirements.

At this step, some prominent questions that need answering can be summarized as:

- *What is the physics that is to be modeled?*
- *What is the scenario for which the physics is modeled??*
- *How many disciplines are involved?*
- *To which detail level the physics has to be modeled?*

3. Establish value objectives:

Value of a computational tool can be about many factors including but not limited to the ease of access, ease of development, budget and time constraints, required fidelity and coupling options. If there is any previous experience or documented results regarding the tool, they can also be important decision factors. Especially any guarantees on the prediction uncertainty, or the availability of a validation data set can be deciding factors in the case of a safety-critical

application. It is crucial to remember that any scientist should remain skeptic to the outputs of a computational tool in the absence of some kind of a benchmark. Some of the questions need answering at this step are:

- *Are there additional Quantities of Interest that might help with the analyses?*
- *Is there previous experience regarding the tool?*
- *Is there available validation data to establish trust in these tools?*
- *How hard is it to develop the tool?*
- *Are there any license-related restrictions?*

4. Generate feasible alternatives:

When the list of available tools and methods for each discipline is compiled, the engineers will have a number of tools that *can* be used for their purpose. However, there are many factors as to some tools or a combination of them can be incompatible, in which case they should be discarded. A tool can be inaccessible, too expensive, hard to learn, or black-box. Some value objectives defined in the previous step may render tools infeasible.

One of the main considerations is the tool fidelity. Engineers should analyze the capabilities of each tool, and try to understand if the tool is capable enough to model the physics in question accurately. However, for early-design applications, too much fidelity can also be an issue. As mentioned before, these tools require so much information about the artifact to produce accurate results. Such information simply may not be available in the current point of the design cycle.

Another challenging aspect for multi-disciplinary problems is the framework of coupling different tools. This must be investigated meticulously, in order to avoid any pitfalls later on. There is often more than one way to connect different disciplinary tools. Generally speaking, the output format of most tools is given in the documentation, and engineers need to confirm that these data formats are compatible with other tools in the loop. Another potential pitfall is the increase in computational time due to coupling of different analyses, in which case coupling multiple higher-fidelity models are generally avoided.

5. Evaluate alternatives:

With the end of the previous step, there should be a list of tools, their capabilities, pros/cons, benchmark/validation data sets, or any other metric that the resulting modeling and simulation environment will be evaluated on. Every entity and every project will have a different set of evaluation criteria. This evaluation may be performed among stakeholders, analysts, subject matter experts etc.

6. Make a decision:

The final step is to select a subset of tools based on previously defined evaluation criteria among the alternatives. Because the simulation environment has not been set up yet, some practitioners may opt to identify other alternatives as safety nets in case the first selection does not turn out to be the optimal one as predicted.. Or simply in terms of fidelity, more than one tool can be utilized in conjunction, as an ensemble of multi-fidelity models in the same or different stages of the product life cycle.

III. Impact of Model Selection on System-Level Critical Uncertainties in Design

A. Modeling and Simulation Environment Alternatives for Aero-Structures Analysis

As previously discussed in Section II.B.5, design of a new aircraft generally start by using low fidelity models, often based on regressions. These tools include significant model-form uncertainty because of the higher level of abstractions involved. Moreover, because the design is also abstracted as many attributes of design have not been decided yet, there is also significant parameter uncertainty. The motivation of this work was to identify the *critical uncertainties* pertaining to the computational tools used in the design, so that they can later be addressed by targeted *validation experiments*. In order to realize this goal, the framework illustrated in Figure 6 was used. First, the methods described in Section II.B were used to decompose the multi-disciplinary problem. In order to guide the analysis, mission-level critical uncertainties are identified using FLOPS [50], a regression-based, low-fidelity analysis. The next step is to increase the fidelity to obtain more information to reduce the overall epistemic uncertainty, and finally propagate the impacts to the level of consequence. This section discusses the comparison and the impact of two different modeling

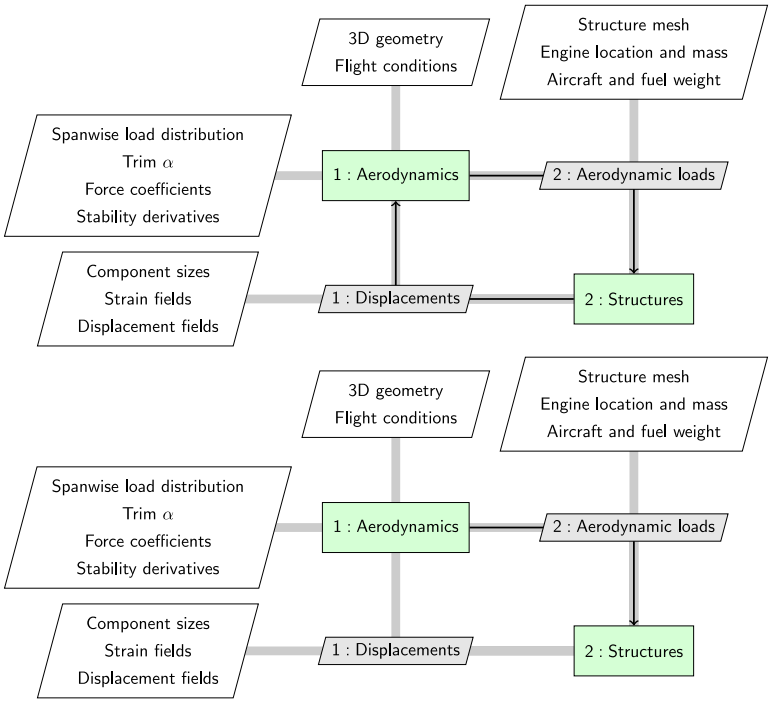


Fig. 14 Two different xDSM approaches for aeroelastic analyses. Decoupled aero-structures (top) and coupled analyses (bottom)

approaches of how to deal with these uncertainties. In order to demonstrate the differences in modeling approach and their implications, a Lockheed Martin C-5M Super Galaxy was selected as the use case because it has some publicly available data to create an aircraft-level mission analysis and wing planform models.

Depending on the purpose of the modeling activity, and the current point in the life cycle, the best choice of the modeling environment may differ. As seen in Figure 13, there is often a number of modeling approaches available for each discipline involved. When the models are selected for each discipline, then appropriate coupling options need to be explored for MDA. For the problem of aero-structures analysis of an aircraft wing in the design phase, two different MDA options are illustrated in Figure 14 in the form of xDSM diagrams. The difference between the top and the bottom diagrams is the addition of a more strict coupling (and any resultant process) of the aerodynamic and structures analysis. The presence of another loop represent an increase in model complexity, and result in additional computational burden. However, in most cases, the resultant analysis should bear more resemblance to the actual physics phenomenon.

More detailed xDSM regarding the two analysis scenarios, including the information pertaining to how the analyses are executed are given in Figures 15 and 16. For these calculations following tools were used: OpenVSP [51] was used for geometry generation, Athena Vortex Lattice [52] for aerodynamic analyses, NASTRAN [53] for structural analyses and HyperSizer [54] for sizing the internal structure thicknesses. Rapid Airframe Design Environment (RADE) [55] was used to wrap these analyses in a single MDA loop. For more discussion on how the modeling environment is set up, please see References [37] and [56].

The Design of Experiment (DoE) operation identified as process #1 in Figure 15 defines uncertainty factors (*u-factors*) in certain uncertain design variables. Although all parameters have a certain uncertainty, the following subset of parameters were selected in order to save computational cost, since they had a higher impact on mission-level metrics [37]. These parameters are: gross and engine weights (W_{TO} and W_{eng}, respectively) and material properties (Young's Modulus E, material density ρ, Poisson Modulus ν) of the aircraft structures. In order to avoid any unnecessary bias, they are sampled from a uniform probability distribution. Next, an aerodynamics analysis (process #3 pink rectangle labelled Aerodynamics) calculates the aerodynamic coefficients of the aircraft during a specified flight condition using a given outer mold line (OML). The aerodynamic coefficients are used to compute the loads on the aircraft, which produces the load distribution needed to calculate the structural deformation. Because the impact of the uncertainty factors are included in the analysis, the resulting load distributions are uncertain.

After the aerodynamic analyses are performed and aerodynamic loads are generated, they are passed to the structural analysis tool. However, because the structural element thicknesses in the wing are not known, a sizing loop must be performed to estimate a realistic internal structure. In other words, the thicknesses of the wing's ribs and spars, determined by the aircraft's need to withstand critical loads, will be used to calculate the wing box weight. The sizing loop is a representation of a design process for the wing's internal structure and is illustrated by processes 5–8 in Figure 15. In the sizing loop, initial guesses of the internal structure thicknesses are given to conduct an initial structural analysis that will output internal loads to inform a structural sizing tool. The sizing loop will continue until the thicknesses of the internal loads converge. Finally, this workflow will output wing box weight, which will have of the uncertainty introduced in aerodynamic and structural analyses. It is important to remember that the purpose of the process is to quantify the impact of parameter uncertainties in the models and not design optimization of the structure for a given mission. Therefore, traditional design parameters such as wing aspect ratio, thickness, taper, sweep, dihedral, and twist are not included in the study.

The xDSM for the coupled aerostructural analysis is illustrated in Figure 16, and it follows a very similar procedure to the previous one. However, there is an additional MDA loop includes processes numbered 5-9. Although this makes the run time of the coupled aerostructural analysis to be significantly longer, it is intended to be a better representation of the physical problem. In this loop, the displacements obtained from the structural analyses are fed back to the aerodynamic analysis to modify the geometry used in aerodynamic load calculations at every step. Consequently, this extra loop requires more function calls, but it helps generate more consistent results between the structural and aerodynamic analyses.

B. Results

In this section, the results for the aero-structures analyses at the wing-level and their implications at the aircraft level are briefly discussed. These results were initially published in an earlier work [37], some of them are repeated here for convenience to the reader. Figure 17 shows the deflection of the wing box under two different aerodynamic loading conditions that occur when $n = -1g$ and $n = 2.25g$, for the coupled and uncoupled MDA workflows. The difference in wing deformation is visible between the two analysis types, as expected. The change in local angle of attack results in increases in aerodynamic loading, hence more deformation. These results highlight how the set of modeling assumptions and the tool capabilities impact the results. The engineers should pay a close attention to *what* the results they obtain mean, especially when they are analyzing critical cases on the mission envelope.

In order to find the implications of uncertainties at different levels, sensitivity analyses need to be carried out. Using the results from both MDA runs, two different surrogate models (polynomial regressions with 5th order terms) were created and used to perform sensitivity analyses. In this study, the points are sampled from uniform probability distributions assumed on input variables. The results given in the prediction profiler in Figure 18 were sampled from these surrogate models.

For uncertainty quantification purposes, looking at general statistics such as mean or standard deviation is a good start, but it does not tell the whole story. Because the uncertain parameters are sampled from distributions, the change in posterior distributions with regards to the input parameters can also be analyzed. A *prediction profiler* that shows the relative importance of six uncertain input variables on wing-level outputs is given in Figure 18. The slopes in each square represent the sensitivity of the metric given in the y-axis to the input parameter in the corresponding x-axis. The blue lines and the distributions represent the uncoupled MDA framework, whereas the orange ones, the coupled MDA. The bands around the plots represent the model-fit uncertainty pertaining to the surrogate model, not model-form or parameter uncertainties. In this figure, it is seen that not only is there a shift in the distribution, the overall shape of the distribution also change with respect to the analysis type and the corresponding set of assumptions and solution mechanisms. Such changes will be more prominent if there is a starker difference between the analysis methods. However, looking at the output distributions given in the right-most column, it is hard to say that a stringent set of modeling assumptions—or a coupling mechanism—will result in tighter distributions, or will converge to a deterministic value for all parameters. However, it is clearly seen that parameters related to physical components playing more part in the bending modes were impacted more by the change in the analysis method, whereas the distribution in skin related parameters is virtually unchanged.

Then, the same surrogate models were used to propagate the results obtained in wing-level analyses to the aircraft-level analysis tool, FLOPS. At aircraft-level, the difference between the two MDA frameworks is still seen to be distinct [37]. A change in the modeling and simulation environment impacts the probability distributions describing the likelihood that the designed aircraft will meet prescribed requirements. Since this will undoubtedly impact the decision making,

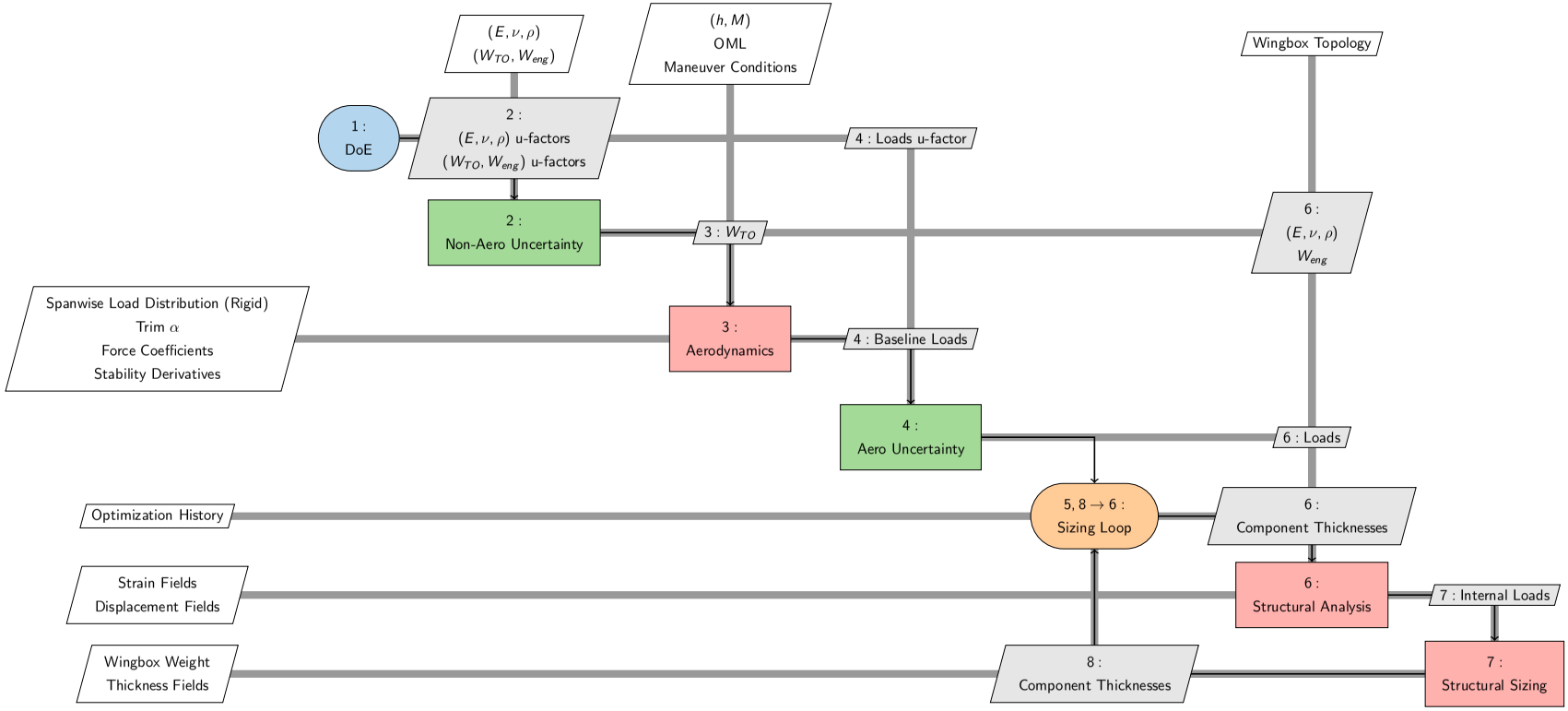


Fig. 15 xDSM for the uncoupled aerostructures analysis

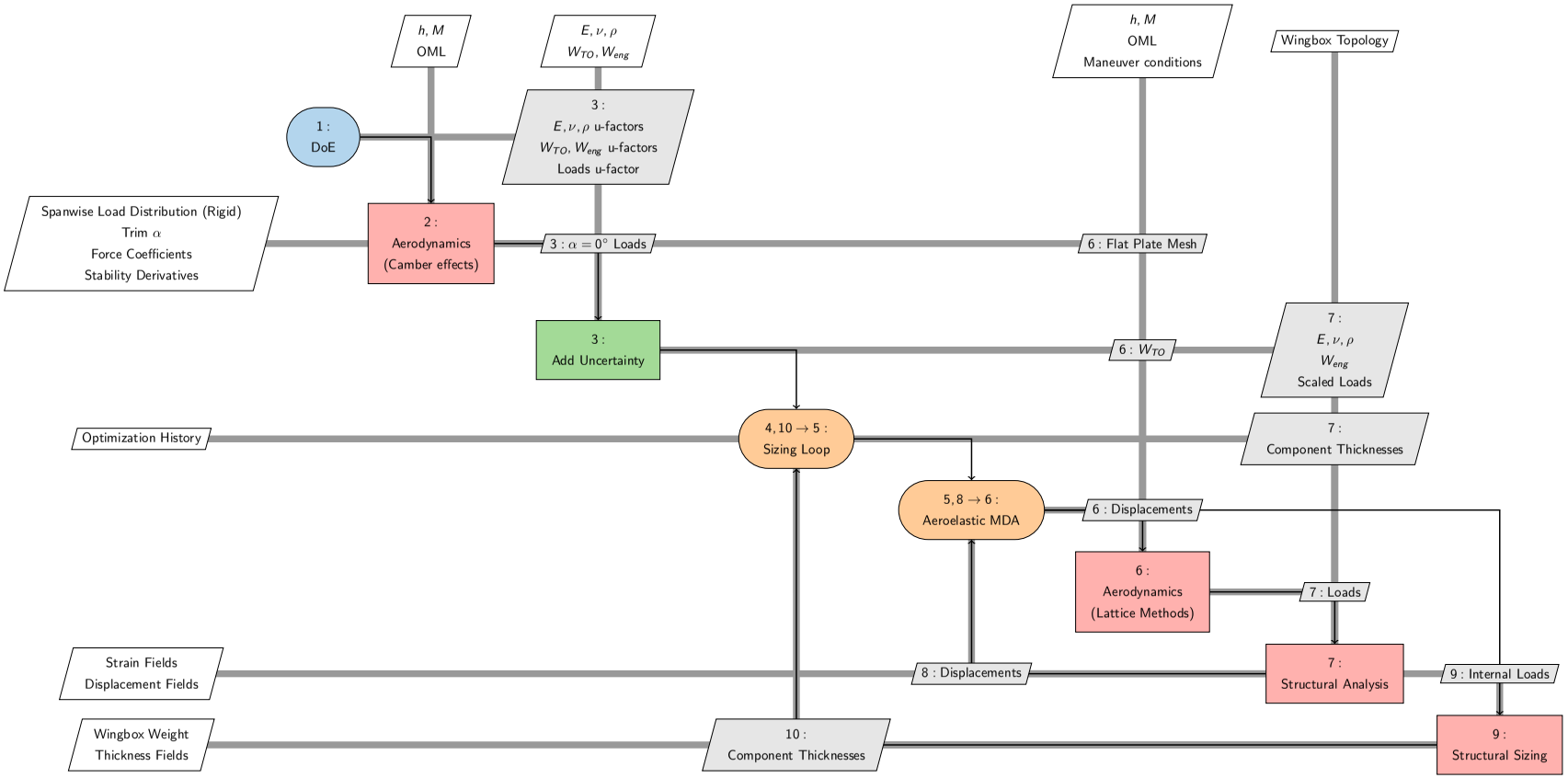


Fig. 16 xDSM for the coupled aerostructures analysis

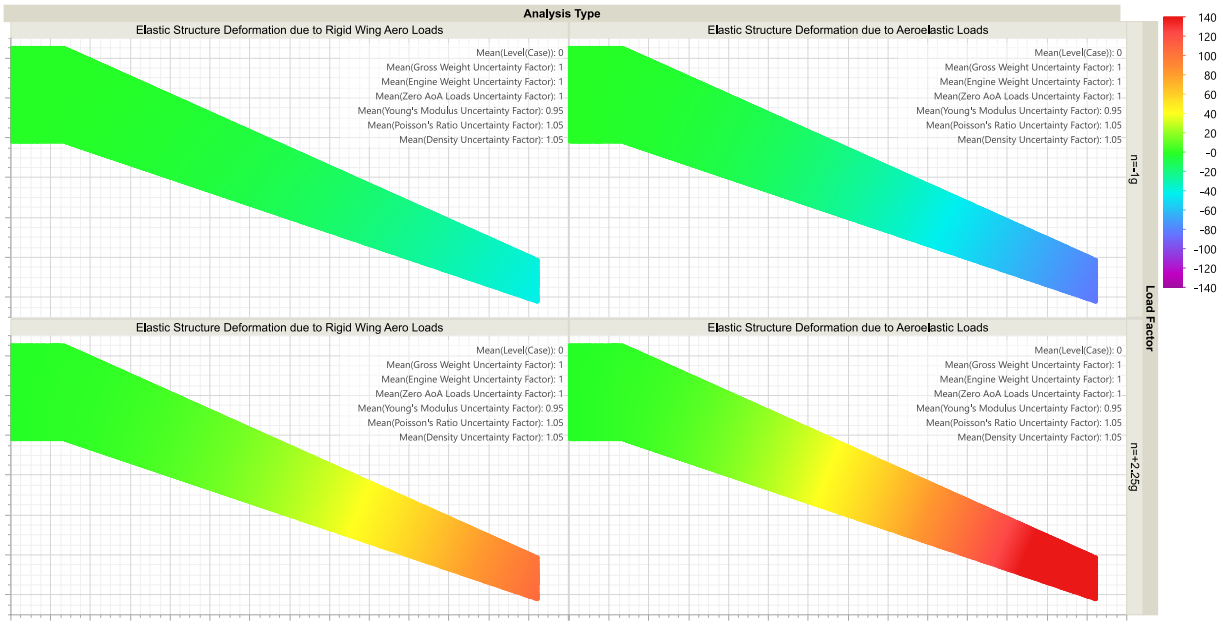


Fig. 17 Wing box deflection due to aerodynamic loads Top row: Load Factor = –1g, Bottom row: Load Factor = +2.25g. Left column: Decoupled Analysis, Right column: Coupled Analysis

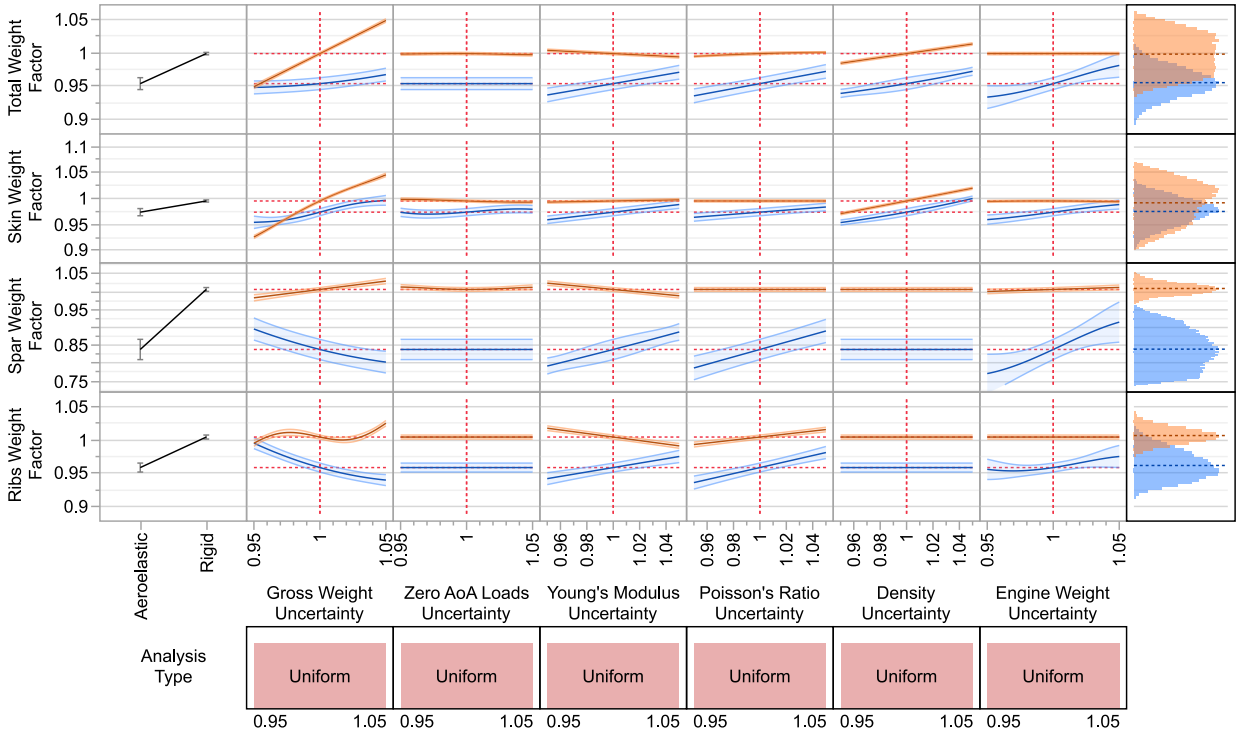


Fig. 18 Prediction profiler for wing-level variables (decoupled analysis in orange, coupled analysis in blue) [37]

Decoupled Aerostructures	Range Factor Contribution	Portion	Rank
Gross Weight Uncertainty Factor	0.059754	0.7822	1
Density Uncertainty Factor	0.008063	0.1056	2
Engine Weight Uncertainty Factor	0.002853	0.0373	3
Young's Modulus Uncertainty Factor	0.002333	0.0305	4
Zero AoA Loads Uncertainty Factor	0.002144	0.0281	5
Poisson's Ratio Uncertainty Factor	0.001242	0.0163	6

Coupled Aeroelasticity	Range Factor Contribution	Portion	Rank
Density Uncertainty Factor	0.014014	0.3766	1
Young's Modulus Uncertainty Factor	0.009868	0.2652	2
Gross Weight Uncertainty Factor	0.006897	0.1853	3
Poisson's Ratio Uncertainty Factor	0.003245	0.0872	4
Engine Weight Uncertainty Factor	0.001659	0.0446	5
Zero AoA Loads Uncertainty Factor	0.001528	0.0411	6

Fig. 19 Critical sensitivities for the aircraft range. Top: Decoupled analysis, Bottom: Coupled analysis [37]

the engineers should be very careful of the modeling assumptions they are working under, especially for the early-design studies.

Figure 19 shows the results of the sensitivity analysis carried the aircraft range for both frameworks. Although it is only one of the mission-level parameters, it is arguable one of the most important ones for a military mobility transport aircraft. It can be seen that, when the aero-structures analyses are not coupled, the sensitivity on take-off gross weight is drastically more compared to the other variables. However, when there is a tight coupling between the two, it overall impact decreases to 18.5%, down from 78%, whereas the impact of parameters that play an important role in the coupling mechanism increase. Especially the material density ρ and the Young's Modulus E become the dominant factors to which the aircraft range—a mission-level parameter—is sensitive.

IV. Conclusion

In this work, established systems-engineering practices are employed for development and selection of a modeling and simulation environment for aero-structures analysis of an aircraft wing. A C-5M Super Galaxy was selected as a use case. The ontology of the problem is described by decomposing the requirements, design mission, functions and the physical architecture. Then, the information and requirements obtained from these decompositions are mapped onto the *modeling architecture*. It is the distinctive aspect of the proposed approach and it contains information about the set of models that can be used to model the reality of interest, such as inputs, outputs, physics modeling capabilities, sets of assumptions and abstractions, coupling strategies run time etc. It is acknowledged that formulating a problem in the proposed fashion the can be quite tedious and maybe even daunting. However, following a systematic approach and having a detailed and accurate modeling decomposition will allow all parties involved in the process to be aware of the state and limitations of the modeling environments used, as well as enabling traceability throughout the process. Mapping of the requirements on to candidate computational tools enables verification for fitness-for-purpose for the given point in the product life cycle. The selection of appropriate level of fidelity and level of abstraction will ensure that the simulation environment is indeed able to yield satisfactory predictions about the response of the system.

The down-selection from a set of candidate models and different coupling options is another step that the engineers should be very careful of. The impact on having two different coupling options for the same problem is demonstrated by comparing the propagated uncertainty in six design variables through two different MDA workflows, where one is a uncoupled aero-structures analysis, and the other is coupled. Not only the change in contributions of the simulation parameters on higher-level metrics is significant, also their ordering is different. Furthermore, the change in the distribution shapes on the propagated variables was observed, as well as the change in statistics. It can be argued that such changes can affect the decision making based on simulation results. Therefore, it is crucial that all parties involved in this process are aware of their goals, the capabilities and limitations of their simulation environment, their goals and the used solution approach. Existence of ambiguity and discord in this process may result in performance shortcomings, schedule delays, or even catastrophic failure.

This framework should be carried out before any simulation runs. It is only natural that one learns new information

about model at hand, or finds about its limitations. In such cases, where the selected model is deemed inappropriate, the decompositions will need to be updated so that a new selection can be made. Furthermore, when the results are collected, they can be scrutinized with the set of assumptions of the simulation environment to serve as an additional step for verifying that obtained results are not obviously inaccurate. This will be an important check point in applications where appropriate validation data does not exist. It also speaks to how the proposed approach is dynamic. For different products or different points in the life cycle, the involved components can be updated with new or required knowledge. The sole purpose of the proposed methodology is to present a systematic way of model selection or development to provide traceability and enable verification of fitness-for-purpose for complex system simulations.

Acknowledgments

The authors acknowledge valuable input provided by Joseph H. Morrison (NASA), Eric Walker (NASA), and especially Nigel Taylor (MBDA) throughout the development of the proposed methodology. The decompositions provided in Section II, are discussed in detail and length, and how they can be used from a utilitarian perspective in Reference 57.

References

[1] Department of Defense, "MIL-STD-1530DC1 Aircraft Structural Integrity Program (ASIP)," , 2016. URL `http://www.everyspec.com`.

[2] Karlberg, M., Löfstrand, M., Sandberg, S., and Lundin, M., "State of the art in simulation-driven design," *International Journal of Product Development*, Vol. 18, 2013, pp. 68–87. https://doi.org/10.1504/IJPD.2013.052166, URL https://www.researchgate.net/publication/264836216.

[3] Olivares, G., Acosta, J. F., and Yadav, V., "Certification of Analysis I & II," *FAA JAMS Technical Review Meeting*, 2010.

[4] Song, K., Lewe, J. H., and Mavris, D. N., "Reenacting the history of the U.S. air transportation network evolution," *Aviation Technology, Integration, and Operations Conference*, 2018, pp. 1–12. https://doi.org/10.2514/6.2018-4240.

[5] Cianetti, F., Morettini, G., Palmieri, M., and Zucca, G., "Virtual qualification of aircraft parts: test simulation or acceptable evidence?" *Procedia Structural Integrity*, Vol. 24, 2019. https://doi.org/10.1016/j.prostr.2020.02.047, URL www.sciencedirect.com.

[6] Seshadri, B. R., and Krishnamurthy, T., "Structural Health Management of Damaged Aircraft Structures Using the Digital Twin Concept," *AIAA SciTech Forum*, 2017. https://doi.org/10.2514/6.2017-1675, URL http://arc.aiaa.org.

[7] Aydemir, H., Zengin, U., Durak, U., and Hartmann, S., "The Digital Twin Paradigm for Aircraft-Review and Outlook," *AIAA SciTech Forum*, 2020. https://doi.org/10.2514/6.2020-0553, URL http://arc.aiaa.org.

[8] Oberkampf, W. L., and Roy, C. J., *Verification and Validation in Scientific Computing*, Cambridge University Press, 2010.

[9] Roy, C. J., and Oberkampf, W. L., "A Complete Framework for Verification, Validation, and Uncertainty Quantification in Scientific Computing (Invited)," *48th AIAA Aerospace Sciences Meeting Including the New Horizons Forum and Aerospace Exposition*, 2010. https://doi.org/10.2514/6.2010-124, URL http://arc.aiaa.org.

[10] Box, G., "Robustness in the Strategy of Scientific Model Building," , 1 1979. https://doi.org/10.1016/B978-0-12-438150-6.50018-2.

[11] Price, M., Raghunathan, S., and Curran, R., "An integrated systems engineering approach to aircraft design," *Progress in Aerospace Sciences*, Vol. 42, 2006, pp. 331–376. https://doi.org/10.1016/J.PAEROSCI.2006.11.002, URL www.elsevier.com/locate/paerosci.

[12] Nolan, D. C., Tierney, C. M., Armstrong, C. G., and Robinson, T. T., "Defining Simulation Intent," *Computer-Aided Design*, Vol. 59, 2015, pp. 50–63. https://doi.org/10.1016/J.CAD.2014.08.030.

[13] Thunnissen, D. P., "Propagating and Mitigating Uncertainty in the Design of Complex Multidisciplinary Systems," , 2005.

[14] Gatian, K. N., "A Quantitative, Model-Driven Approach To Technology Selection and Development Through Epistemic Uncertainty Reduction," , 2015.

[15] Oberkampf, W. L., DeLand, S. M., Rutherford, B. M., Diegert, K. V., and Alvin, K. F., "Error and uncertainty in modeling and simulation," *Reliability Engineering and System Safety*, Vol. 75, 2002, pp. 333–357. https://doi.org/10.1016/S0951-8320(01)00120-X, URL www.elsevier.com/locate/ress.

[16] Najm, H. N., "Uncertainty Quantification and Polynomial Chaos Techniques in Computational Fluid Dynamics," *Annual review of fluid mechanics*, Vol. 41, 2008, pp. 35–52. https://doi.org/10.1146/ANNUREV.FLUID.010908.165248, URL https://www.annualreviews.org/doi/abs/10.1146/annurev.fluid.010908.165248.

[17] Trucano, T. G., Swiler, L. P., Igusa, T., Oberkampf, W. L., and Pilch, M., "Calibration, validation, and sensitivity analysis: What's what," *Reliability Engineering & System Safety*, Vol. 91, 2006, pp. 1331–1357. https://doi.org/10.1016/J.RESS.2005.11.031.

[18] Sankararaman, S., and Mahadevan, S., "Likelihood-based representation of epistemic uncertainty due to sparse point data and/or interval data," *Reliability Engineering and System Safety*, Vol. 96, 2011, pp. 814–824. https://doi.org/10.1016/j.ress.2011.02.003.

[19] Hacking, I., *The Emergence of Probability: A Philosophical Study of Early Ideas about Probability, Induction and Statistical Inference*, 2nd ed., Cambridge University Press, 1975. https://doi.org/10.1111/j.1468-0149.1976.tb01651.x.

[20] Prtija, S. S., "Alea iacta est: Caesar's Crossing of the Rubicon," *Sineza*, Vol. 1, 2020, pp. 31–45.

[21] Yao, W., Chen, X., Luo, W., Tooren, M. V., and Guo, J., "Review of uncertainty-based multidisciplinary design optimization methods for aerospace vehicles," *Progress in Aerospace Sciences*, Vol. 47, 2011, pp. 450–479. https://doi.org/10.1016/J.PAEROSCI.2011.05.001.

[22] Thunnissen, D. P., "Uncertainty Classification for the Design and Development of Complex Systems," *3rd Annual Predictive Methods Conference*, 2003, p. 16. URL http://citeseerx.ist.psu.edu/viewdoc/download?doi=10.1.1.128.133&rep=rep1&type=pdf.

[23] Kroll, N., Abu-Zurayk, M., Dimitrov, D., Franz, T., Führer, T., Gerhold, T., Görtz, S., Heinrich, R., Ilic, C., Jepsen, J., Jägersküpper, J., Kruse, M., Krumbein, A., Langer, S., Liu, D., Liepelt, R., Reimer, L., Ritter, M., Schwöppe, A., Scherer, J., Spiering, F., Thormann, R., Togiti, V., Vollmer, D., and Wendisch, J. H., "DLR project Digital-X: towards virtual aircraft design and flight testing based on high-fidelity methods," *CEAS Aeronautical Journal*, Vol. 7, 2016, pp. 3–27. https://doi.org/10.1007/s13272-015-0179-7, URL https://link.springer.com/article/10.1007/s13272-015-0179-7.

[24] Collar, A. R., "The Expanding Domain of Aeroelasticity," *The Journal of the Royal Aeronautical Society*, Vol. 50, 1946, pp. 613–636. https://doi.org/10.1017/S0368393100120358, URL https://www.cambridge.org/core/product/identifier/S0368393100120358/type/journal_article.

[25] Stern, F., Wilson, R., and Shao, J., "Quantitative V&V of CFD simulations and certification of CFD codes," *International Journal for Numerical Methods in Fluids*, Vol. 50, 2006, pp. 1335–1355. https://doi.org/10.1002/FLD.1090, URL https://onlinelibrary.wiley.com/doi/full/10.1002/fld.1090https://onlinelibrary.wiley.com/doi/abs/10.1002/fld.1090https://onlinelibrary.wiley.com/doi/10.1002/fld.1090.

[26] Xie, J., Cai, Y., Chen, M., and Mavris, D. N., "A Multi-Fidelity Aerodynamic Analysis Method for Transonic Transport Aircraft Conceptual Design and Mission Analysis," *AIAA Aviation*, 2021. https://doi.org/10.2514/6.2021-2433, URL https://www.researchgate.net/publication/353537455.

[27] Rumpfkeil, M. P., and Beran, P., "Multi-fidelity surrogate models for flutter database generation," *Computers and Fluids*, Vol. 197, 2020, p. 104372. https://doi.org/10.1016/j.compfluid.2019.104372, URL https://doi.org/10.1016/j.compfluid.2019.104372.

[28] Yong, H. K., Wang, L., Toal, D. J., Keane, A. J., and Stanley, F., "Multi-fidelity Kriging-assisted structural optimization of whole engine models employing medial meshes," *Structural and Multidisciplinary Optimization*, Vol. 60, 2019, pp. 1209–1226. https://doi.org/10.1007/S00158-019-02242-6/FIGURES/12, URL https://link.springer.com/article/10.1007/s00158-019-02242-6.

[29] Diazdelao, F. A., and Adhikari, S., "Bayesian assimilation of multi-fidelity finite element models," *Computers & Structures*, Vol. 92-93, 2012, pp. 206–215. https://doi.org/10.1016/J.COMPSTRUC.2011.11.002.

[30] Mittal, S., and Martin, J. L. R., "Model-driven systems engineering for netcentric system of systems with DEVS unified process," *Proceedings of the 2013 Winter Simulation Conference - Simulation: Making Decisions in a Complex World, WSC 2013*, IEEE, 2013, pp. 1140–1151. https://doi.org/10.1109/WSC.2013.6721503.

[31] Zimmerman, P., "MBSE in the Department of Defense," , 5 2015. URL https://ses.gsfc.nasa.gov/ses_data_2015/150512_Zimmerman.pdf.

[32] Hatakeyama, S. J., Seal, D. W., Farr, D., and Haase, S. C., "An Alternate View of the Systems Engineering "V" in a Model-Based Engineering Environment," *AIAA Space and Astronautics Forum and Exposition*, 2018. https://doi.org/10.2514/6.2018-5326, URL http://arc.aiaa.org.

[33] Allouis, E., Blake, R., Gunes-Lancet, S., Jorden, T., Maddison, B., Schroeven-Deceuninck, H., Stuttard, M., Truss, P., Ward, K., Ward, R., and Woods, M., "A Facilitiy for the Verification and Validation of Robotics and Autonomy for Planetary Exploration," *Proceedings of the 12th Symposium on Advanced Space Technologies in Automation and Robotics.*, 2013. URL https://www.researchgate.net/publication/320585817.

[34] Balci, O., "Quality assessment, verification, and validation of modeling and simulation applications," *Proceedings - Winter Simulation Conference*, Vol. 1, IEEE, 2004, pp. 122–129. https://doi.org/10.1109/wsc.2004.1371309, simulation requirements.

[35] Balci, O., "Verification, Validation, and Certification of Modeling and Simulation Applications," *Proceedings of the 2003 Winter Simulation Conference*, edited by S. Chick, P. Sanchez, D. Ferrin, and M. DJ, 2003. URL www.praxiom.com/accrediters.htm.

[36] Lambe, A. B., and Martins, J. R., "Extensions to the design structure matrix for the description of multidisciplinary design, analysis, and optimization processes," *Structural and Multidisciplinary Optimization*, Vol. 46, 2012, pp. 273–284. https://doi.org/10.1007/S00158-012-0763-Y/FIGURES/12, URL https://link.springer.com/article/10.1007/s00158-012-0763-y.

[37] Yarbasi, E., Bagdatli, B., and Mavris, D., "System-Level Identification of Critical Uncertainties to Enable Validation Experiments," *33rd Congress of the International Council of the Aeronautical Sciences*, 2022.

[38] Mavris, D. N., Delaurentis, D. A., Bandte, O., and Hale, M. A., "A stochastic approach to multi-disciplinary aircraft analysis and design," *36th AIAA Aerospace Sciences Meeting and Exhibit*, AIAA, 1998. https://doi.org/10.2514/6.1998-912.

[39] Guarino, N., Oberle, D., and Staab, S., "What Is an Ontology?" , 2009. https://doi.org/10.1007/978-3-540-92673-3_0, URL https://link.springer.com/chapter/10.1007/978-3-540-92673-3_0.

[40] ISO/IEC/IEEE, "15939-2017 - ISO/IEC/IEEE International Standard - Systems and software engineering–Vocabulary," , 2017.

[41] Romli, F., "Functional Analysis for Conceptual Aircraft Design," *Journal of Advanced Management Science*, Vol. 1, 2013. https://doi.org/10.12720/joams.1.4.349-353.

[42] International Organization for Standardization, "ISO - ISO/IEC/IEEE 24748-1:2018 - Systems and software engineering — Life cycle management — Part 1: Guidelines for life cycle management," , 2010. URL https://www.iso.org/standard/72896.html.

[43] de Sousa Cardeira, A., "Aeroelastic Analysis of Aircraft Wings," , 2014.

[44] Duignan, B., Fumerton, R., Quinton, A. M., and Quinton, B., "Encyclopedia Britannica," , 2020. URL https://www.britannica.com/topic/empiricism.

[45] Afonso, F., Vale, J., Éder Oliveira, Lau, F., and Suleman, A., "A review on non-linear aeroelasticity of high aspect-ratio wings," *Progress in Aerospace Sciences*, Vol. 89, 2017, pp. 40–57. https://doi.org/10.1016/J.PAEROSCI.2016.12.004.

[46] Pilch, M., Trucano, T., Moya, J., Froehlich, G., Hodges, A., and Peercy, D., "Guidelines for Sandia ASCI Verification and Validation Plans-Content and Format: Version 2.0," , 2001.

[47] Diamond, D. J., "Experience Using Phenomena Identification and Ranking Technique (PIRT) for Nuclear Analysis," , 10 2006. URL https://www.bnl.gov/isd/documents/32315.pdf.

[48] Schrage, D. P., and Mavris, D. N., "Integrated Product/Process Design/Development (IPPD) Through Robust Design Simulation: The Key for Affordable Systems," *AIAA Aircraft Engineering, Technology and Operations Congress*, AIAA, 1995. https://doi.org/10.2514/6.1995-3892, URL http://arc.aiaa.org.

[49] Oberkampf, W., "What Are Validation Experiments?" *Experimental Techniques*, Vol. 25, 2001, pp. 35–40. https://doi.org/10.1111/j.1747-1567.2001.tb00023.x.

[50] Wells, D. P., Horvath, B. L., and McCullers, L. A., "The flight optimization system weights estimation method," Tech. rep., 2017.

[51] McDonald, R. A., and Gloudemans, J. R., "Open Vehicle Sketch Pad: An Open Source Parametric Geometry and Analysis Tool for Conceptual Aircraft Design," *AIAA SciTech Forum*, 2022. https://doi.org/10.2514/6.2022-0004, URL https://arc.aiaa.org/doi/abs/10.2514/6.2022-0004.

[52] Drela, M., and Youngren, H., "Athena Vortex Lattice," Electronic, Jun. 2022. URL https://web.mit.edu/drela/Public/web/avl/.

[53] MSC Software, "MSC NASTRAN," Electronic, Jun. 2022. URL https://www.mscsoftware.com/product/msc-nastran.

[54] Collier Aerospace, "HyperSizer," Electronic, Jun. 2022. URL https://hypersizer.com/.

[55] Corman, J. A., Weston, N. R., Friedland, C. I., Mavris, D. N., and Laughlin, T. W., "Rapid Airframe Design Environment (RADE): A Parametric, Modular, and Multidisciplinary Framework for Conceptual Phase Airframe Design," *2018 AIAA Modeling and Simulation Technologies Conference*, 2018. https://doi.org/10.2514/6.2018-1930, URL http://arc.aiaa.org.

[56] Yarbasi, E. Y., Bagdatli, B., and Mavris, D., "Demonstration of the Framework on a Military Mobility Transport," *NATO AVT-297 Report*, 2023.

[57] Taylor, N., Mavris, D., Yarbasi, E. Y., and Bagdatli, B., "A Process Framework for Identifying Requirements for Physical Referent Data to Support Computational Model Validation," *AIAA SciTech Forum*, AIAA, 2023.